

Soil Physics Workshop

Soil Hydrology with the Stevens HydraProbe Soil Sensor

Time:	TBD
Date:	TBD
Length:	2 to 3 hours
Instructor:	Keith Bellingham, P.H. Hydrologist/Soil Scientist for Stevens Water Monitoring Systems



Course Summary

Knowledge of soil moisture and the measurement of soil moisture help society answer some important questions about drought, flood forecasting, greenhouse gas emissions, food security and crop production. This course will focus on the science of the analytical measurements of soil water content and water potential. From an applications in the field perspective to a more theoretical approach, this course provides tools, information, and insight for users of soil moisture sensors and soil moisture data.



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 **HydraProbe**

Course Syllabus

1) History / About Stevens Water Monitoring (20 mins)

- History of Stevens, founded in 1911
- Our mission to serve the customer
- Communication products update
- Sensors and accessories

2) Soil Geomorphology (10 mins)

- Soil taxonomy
- Soil development
- Horizons and vertical profiles
- Global soil orders

3) HydraProbe Introduction (20 mins)

- The broad topic of soil sensors
- The development of the HydraProbe

4) Soil Sensor Applications (20 mins)

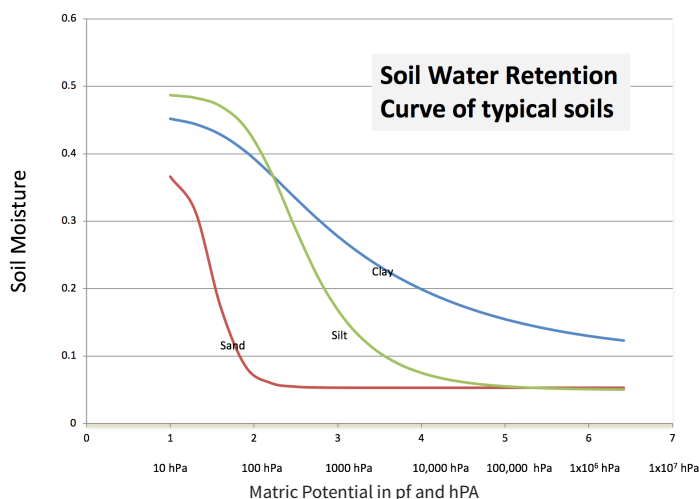
- Movement of water in the vadose zone
- The significance of field capacity to an irrigator
- Where are soil sensors used?
- National mesonets and regional networks
- Soil survey
- Soil moisture deficit index as a modeling parameter

5) Soil Water Movement (20 mins)

- Capillary forces
- Field capacity
- Ways to measure matric potential
- Water retention curves
- The Richards equation
- Videos of soil water movement
- Free tools from the USDA

6) Dielectric Permittivity (how does a soil sensor work?) (20 mins)

- Capacitance
- Dipole moments
- Electromagnetic waves
- Dielectric permittivity as a complex number
- The physics of complex impedance
- The advantages of separating the real from the imaginary permittivity
- HydraProbe measurement method



7) Types of Soil Moisture Sensors and how they work (10 mins)

- TDR, TDT, FDR, capacitance sensors, impedance based sensors

8) Calibration of Soil Sensors (20 mins)

- A general calibration
- Validate before you calibrate
- Calibration forms

9) Bulk EC and Salinity (10 mins)

- Why is EC important?
- Pore water EC and how to get it

10) Installation and Troubleshooting (20 mins)

- Before you go into the field
- Preparing the pit
- Wiring and enclosures, what is SDI-12?
- General troubleshooting

11) How to Interpret Soil Moisture Data (10 mins)

- The soil hydrograph
- Dry down and wet up daily and seasonal
- The wetting front

12) Lightning Protection (10 mins)

13) Q&A